BADR UNIVERSITY IN CAIRO (BUC) — FACULTY OF ENGINEERING

Exercise Sheet (6): Hydraulic Accessories — Full Step-by-Step Solutions & Teaching Guide

Course: MEC 457 / MEC 431 — Fluid Power Systems & Control | **Examiner:** Dr. Ziad Ahmed Ibrahim
Teaching Assistant: Eng. Moh Zedan | **Mechatronics & Energy Departments** — Academic Year 2026/2027

Question 1: Principles of Operation and Possible Applications of Hydraulic Accumulators

1.1 Working Principle of Oleo-Pneumatic Accumulators

Because hydraulic liquids are almost incompressible (compressing only about 0.5–1% under 100 bar), they cannot store potential pressure energy directly within practical container dimensions. Gas, on the other hand, is extremely compressible.

> An **oleo-pneumatic accumulator** utilizes the high compressibility of an inert gas (precharged Dry Nitrogen, N_2) separated from the working hydraulic liquid by an elastic physical barrier (a synthetic rubber bladder, a floating piston with elastomeric seals, or a flexible diaphragm).
>
> - **Charging Phase:** When pump delivery pressure exceeds gas precharge pressure, pressurized oil forces its way into the accumulator vessel, compressing the nitrogen gas volume and storing energy.
> - **Holding Phase:** The fluid is held at system pressure ready for instantaneous release without relying on running electric motors.
> - **Discharging Phase:** When system pressure falls below the compressed gas pressure (e.g., during rapid actuator extension), the gas expands, forcing the stored fluid out into the circuit at high velocity.

1.2 Major Industrial Applications

1. **Auxiliary Energy Source (Peak Shaving):** Accommodates short-duration peak flow demands in machines with intermittent duty cycles (e.g., injection molding machines, die-casting presses). The pump can be sized for the low *cycle-average flow* rather than the enormous *peak flow*, cutting pump displacement and electric motor size by 50–70%.
2. **Emergency Power Supply (Fail-Safe Storage):** Acts as a hydraulic battery that holds sufficient pressurized oil to safely bring actuators into a safe home position (closing blowout preventers on oil rigs, raising landing gear, or holding brake pressure) in the event of an electrical blackout.
3. **Leakage and Pressure Compensation:** Maintains clamping pressure on presses or machine tool fixtures over prolonged holding cycles. The pump unloads to tank or shuts off completely, eliminating oil overheating and saving power.
4. **Hydraulic Shock / Water Hammer Suppression:** Placed near fast-acting directional valves to absorb rapid pressure surges (up to 400 bar) caused by sudden deceleration of high-inertia loads.
5. **Pump Pulsation Dampening:** Installed directly at the pump outlet port to smooth out pressure and flow pulsations inherent to gear, vane, and piston pumps, reducing noise and pipe fatigue.
6. **Hydraulic Spring / Vehicle Suspension:** Acts as an adjustable hydro-pneumatic spring on heavy earthmoving machinery, tractors, and mobile cranes.

Question 2: Volumetric Capacity Definition and General Sizing Derivation

2.1 Definition of Volumetric Capacity

The **effective volumetric capacity (ΔV)** of an oleo-pneumatic accumulator is defined as the volume of usable hydraulic fluid that the accumulator can accept and subsequently discharge back into the hydraulic circuit between the maximum system operating pressure (p_2) and the minimum allowable operating pressure (p_1).

2.2 General Thermodynamic Derivation

Consider an accumulator of total internal shell volume V_0 precharged with nitrogen to gas pressure p_0 when completely empty of oil.

State 0: Precharge State (Empty of Oil):
The bladder/diaphragm/piston completely expands to fill the entire shell volume V_0:

$$\text{Gas Pressure} = p_0, \qquad \text{Gas Volume} = V_0, \qquad \text{Oil Volume} = 0$$

State 1: Minimum Operating Pressure (Discharged State):
The system reaches its lowest allowable working pressure p_1. The gas has expanded to volume V_1:

$$\text{Gas Pressure} = p_1, \qquad \text{Gas Volume} = V_1$$

State 2: Maximum Operating Pressure (Fully Charged State):
The system reaches its maximum working pressure p_2 (pump relief setpoint). Oil has compressed the gas to its minimum volume V_2:

$$\text{Gas Pressure} = p_2, \qquad \text{Gas Volume} = V_2$$

Polytropic Gas Expansion Law:
The thermodynamic behavior of the nitrogen gas during compression and expansion is governed by:

$$p \cdot V^n = \text{constant}$$

where:

- $n = 1.0$: **Isothermal Process** (slow charging/discharging over minutes; heat is dissipated through shell walls).
- $n = \kappa = 1.4$: **Adiabatic Process** (rapid discharge over a few seconds; no time for heat transfer).
- $1.0 < n < 1.4$: **Polytropic Process** (general industrial cycles; typically $n \approx 1.25 - 1.3$).

Expressing Gas Volumes in Terms of V_0 and Operating Pressures:

$$p_0 V_0^n = p_1 V_1^n \implies V_1 = V_0 \left(\frac{p_0}{p_1} \right)^{\frac{1}{n}}$$

$$p_0 V_0^n = p_2 V_2^n \implies V_2 = V_0 \left(\frac{p_0}{p_2} \right)^{\frac{1}{n}}$$

Usable Delivered Fluid Volume (ΔV):
Because oil and gas are mutually enclosed in a rigid shell of constant volume V_0, the change in oil volume delivered between state 2 and state 1 equals the change in gas volume:

$$\Delta V = V_{oil,2} - V_{oil,1} = (V_0 - V_2) - (V_0 - V_1) = V_1 - V_2$$

Substituting expressions for V_1 and V_2:

$$\boxed{\Delta V = V_0 \left[\left(\frac{p_0}{p_1} \right)^{\frac{1}{n}} - \left(\frac{p_0}{p_2} \right)^{\frac{1}{n}} \right]}$$

Master Accumulator Sizing Equation for Shell Size V_0:

$$\boxed{V_0 = \frac{\Delta V}{\left(\frac{p_0}{p_1} \right)^{\frac{1}{n}} - \left(\frac{p_0}{p_2} \right)^{\frac{1}{n}}}}$$

(Note: All pressures p_0, p_1, p_2 must strictly be absolute pressures!).

Questions 3, 4 & 5: Construction & Operation of Accumulator Types

3. Piston-Type Accumulators (Question 3)

Construction: Consists of a honed, high-precision seamless steel cylinder barrel, capped with two threaded or bolted end covers (gas end with charging valve and fluid end with hydraulic port). A lightweight floating metal piston equipped with multiple elastomeric sealing rings (O-rings, PTFE back-up rings, and guide rings) slides axially inside to physically separate the gas from the oil.
Operation: When hydraulic oil pressure exceeds gas pressure, the piston is forced upward, compressing the nitrogen. When fluid demand rises, the gas drives the piston downward, discharging the oil.
Engineering Pros & Cons:

- *Advantages:* Accommodates huge volumetric capacities (over 1000 Liters); can be paired with auxiliary nitrogen gas bottles; handles very high working pressures (up to 1000 bar); tolerates wide temperature extremes.
- *Disadvantages:* Higher seal friction and inertia (slower response time to pressure spikes); prone to seal leakage and bore scoring if fluid is contaminated. Must be mounted vertically.

4. Bladder-Type Accumulators (Question 4)

Construction: Consists of a forged seamless steel pressure shell with a pear-shaped flexible elastomeric bladder (typically Nitrile Butyl Rubber, NBR) suspended inside. The top neck of the bladder is vulcanized to a gas charging valve. At the bottom fluid port, a spring-loaded **poppet valve** is integrated.
Function of the Poppet Valve: Essential safety device. When the accumulator discharges all fluid, the expanding bladder pushes against the poppet, mechanically closing the fluid port. This prevents the bladder from extruding out through the pipe fitting and tearing under pressure!
Operation: Gas is precharged inside the bladder. Oil enters around the bladder, squeezing it uniformly. Rapid response due to virtually zero inertia and zero friction.
Engineering Pros & Cons:

- *Advantages:* Instantaneous response (ideal for pulsation dampening and water hammer); zero mechanical friction; excellent sealing (no sliding dynamic seals); highly contamination tolerant.
- *Disadvantages:* Limited volumetric capacity (typically up to 50 Liters); bladder fatigue failure over millions of cycles; precharge pressure ratio is limited: $p_2/p_0 \leq 4:1$ to prevent bladder overstretching.

5. Diaphragm-Type Accumulators (Question 5)

Construction: Consists of two forged steel hemispherical shells welded or bolted together at their equator, clamping a flexible elastomeric diaphragm across the diameter. A small metal button is molded into the center of the diaphragm to seal the oil port when fully discharged.
Operation: Oil occupies the lower hemisphere; nitrogen occupies the upper hemisphere. The diaphragm flexes upward under oil pressure and downward during discharge.
Engineering Pros & Cons:

- *Advantages:* Extremely compact, lightweight, and low-cost; exceptional response speed; can be mounted in any orientation (horizontal, vertical, inverted).
- *Disadvantages:* Very small capacities (typically 0.1 to 4 Liters); limited pressure ratings (up to 250–350 bar); non-repairable if welded. Commonly used in automotive brake systems, hydro-pneumatic suspensions, and small power packs.

Question 6: Sizing of a Hydraulic Accumulator (Numerical Problem)

Problem Statement:
Calculate the size (V_0) of a hydraulic accumulator necessary to deliver four liters of oil ($\Delta V = 4\ \text{L}$) between pressures of $p_2 = 250\ \text{bar}$ and $p_1 = 150\ \text{bar}$, if the charging pressure is $p_0 = 100\ \text{bar}$ (gauge pressure), assuming an adiabatic compression process ($\kappa = 1.4$).

Step-by-Step Mathematical Solution:

Step 1: Convert Gauge Pressures to Absolute Pressures:

$$p_0 = 100 + 1 = 101\ \text{bar (abs)} = 10.1 \times 10^6\ \text{Pa}$$

$$p_1 = 150 + 1 = 151\ \text{bar (abs)} = 15.1 \times 10^6\ \text{Pa}$$

$$p_2 = 250 + 1 = 251\ \text{bar (abs)} = 25.1 \times 10^6\ \text{Pa}$$

$$\Delta V = 4.0\ \text{L} = 4.0 \times 10^{-3}\ \text{m}^3, \qquad \kappa = 1.4 \implies \frac{1}{\kappa} = 0.714286$$

Step 2: Calculate Pressure Ratios:

$$\left(\frac{p_0}{p_1}\right)^{\frac{1}{\kappa}} = \left(\frac{101}{151}\right)^{0.714286} = (0.668874)^{0.714286} = 0.74971$$

$$\left(\frac{p_0}{p_2}\right)^{\frac{1}{\kappa}} = \left(\frac{101}{251}\right)^{0.714286} = (0.402390)^{0.714286} = 0.52130$$

Step 3: Calculate Denominator Difference:

$$\text{Denominator} = \left(\frac{p_0}{p_1}\right)^{\frac{1}{\kappa}} - \left(\frac{p_0}{p_2}\right)^{\frac{1}{\kappa}} = 0.74971 - 0.52130 = 0.22841$$

Step 4: Compute Required Nominal Accumulator Volume V_0:

$$V_0 = \frac{\Delta V}{0.22841} = \frac{4.0\ \text{L}}{0.22841} = \mathbf{17.51\ \text{Liters}}$$

Final Calculated Size of Accumulator:

Nominal Gas Volume: **V_0 = 17.51 Liters**

*(Engineering Selection: Standard commercial catalog size = **20 Liters shell**).*

Question 7: Complete Duty Cycle & Accumulator Sizing Analysis

Problem Statement:
A hydraulic system operates in a regular operating cycle of $T = 50$ s duration. The flow demand during the operating cycle is given as:

- $t \in [0, 4\ \text{s}]$: Flow rate $Q_1 = 0.2\ \text{L/s}$
- $t \in [4, 10\ \text{s}]$: Flow rate $Q = 0.0\ \text{L/s}$ (Idle)
- $t \in [10, 20\ \text{s}]$: Flow rate $Q_2 = 0.4\ \text{L/s}$
- $t \in [20, 35\ \text{s}]$: Flow rate $Q = 0.0\ \text{L/s}$ (Idle)
- $t \in [35, 45\ \text{s}]$: Flow rate $Q_3 = 0.3\ \text{L/s}$
- $t \in [45, 50\ \text{s}]$: Flow rate $Q = 0.0\ \text{L/s}$ (Idle)

Maximum pump delivery pressure during the cycle is $p_1 = 180\ \text{bar}$. The system uses a fixed displacement pump.
Determine:
a) Required pump flow rate when using the pump ONLY.
b) Required pump flow rate and suitable size of the accumulator if used to compensate for short-duration demands, assuming maximum allowable pressure $p_2 = 250\ \text{bar}$.

Part (a): Sizing for Pump ONLY (No Accumulator)

When no accumulator is installed, the pump must be sized to supply the **absolute peak flow demand** of the entire cycle instantaneously:

$$Q_{pump,only} = Q_{peak} = \max(Q(t)) = 0.4\ \text{L/s}$$

Converting to standard engineering units:

$$Q_{pump,only} = 0.4 \times 60 = \mathbf{24.0\ L/min}$$

Electric motor power required at 180 bar (assuming 100% efficiency):

$$N_{motor,only} = \frac{180\ \text{bar} \times 24\ \text{L/min}}{600} = \mathbf{7.2\ kW}$$

⚠️ **Severe Inefficiency:** The pump runs at full displacement (24 L/min) for 50 seconds, but full flow is needed for only 10 seconds! For 29 seconds of the cycle, flow demand is ZERO, dumping 24 L/min across the relief valve and generating massive heat.

Part (b): Sizing with Hydraulic Accumulator (Peak Shaving)

Step 1: Calculate Total Oil Volume Consumed per Cycle:

$$V_{total} = \sum Q_i \cdot \Delta t_i = (0.2 \times 4) + (0.4 \times 10) + (0.3 \times 10) = 0.8 + 4.0 + 3.0 = \mathbf{7.8\ Liters}$$

Step 2: Calculate Required Continuous Pump Flow Rate:
When using an accumulator, the pump operates continuously throughout the 50-second cycle at the **mean (average) flow rate**:

$$Q_{pump} = Q_{avg} = \frac{V_{total}}{T_{cycle}} = \frac{7.8\ \text{L}}{50\ \text{s}} = \mathbf{0.156\ L/s} = \mathbf{9.36\ L/min}$$

Electric motor power required at 180 bar:

$$N_{motor,acc} = \frac{180 \text{ bar} \times 9.36 \text{ L/min}}{600} = \mathbf{2.81 \text{ kW}}$$

*(Power demand slashed from 7.2 kW down to 2.81 kW — over **60% energy savings!**).*

Step 3: Accumulator Fluid Volume Delivery Analysis (ΔV):

Let us trace the net flow balance $\frac{dV_{acc}}{dt} = Q_{pump} - Q_{demand}$ across each interval:

Time Interval	Duration	Demand	Pump Delivery	Accumulator Flow Rate	Volume Change in Acc.
$0 - 4$ s	4 s	0.200 L/s	0.156 L/s	Discharging (-0.044 L/s)	$-0.044 \times 4 = -0.176$ L
$4 - 10$ s	6 s	0.000 L/s	0.156 L/s	Charging (+0.156 L/s)	$+0.156 \times 6 = +0.936$ L
$10 - 20$ s	10 s	0.400 L/s	0.156 L/s	Discharging (-0.244 L/s)	$-0.244 \times 10 = \mathbf{-2.440}$ L ⚠️
$20 - 35$ s	15 s	0.000 L/s	0.156 L/s	Charging (+0.156 L/s)	$+0.156 \times 15 = +2.340$ L
$35 - 45$ s	10 s	0.300 L/s	0.156 L/s	Discharging (-0.144 L/s)	$-0.144 \times 10 = -1.440$ L
$45 - 50$ s	5 s	0.000 L/s	0.156 L/s	Charging (+0.156 L/s)	$+0.156 \times 5 = +0.780$ L

The most severe discharge deficit occurs in interval 3 ($10 - 20$ s), where the accumulator must supply:

$$\Delta V = (0.4 - 0.156) \times 10 = \mathbf{2.44 \text{ Liters}}$$

Step 4: Accumulator Shell Volume Calculation:

Operating pressures:

- Minimum system working pressure: $p_1 = 180 \text{ bar} = 181 \text{ bar (abs)}$
- Maximum allowable system pressure: $p_2 = 250 \text{ bar} = 251 \text{ bar (abs)}$
- Standard Precharge Rule of Thumb: $p_0 = 0.90 \cdot p_1 = 0.90 \times 180 = 162 \text{ bar (gauge)} = 163 \text{ bar (abs)}$

Applying the adiabatic equation ($\kappa = 1.4$):

$$\left(\frac{p_0}{p_1}\right)^{\frac{1}{\kappa}} = \left(\frac{163}{181}\right)^{0.714286} = 0.92716$$

$$\left(\frac{p_0}{p_2}\right)^{\frac{1}{\kappa}} = \left(\frac{163}{251}\right)^{0.714286} = 0.73315$$

$$\text{Difference} = 0.92716 - 0.73315 = 0.19401$$

$$V_0 = \frac{\Delta V}{0.19401} = \frac{2.44 \text{ L}}{0.19401} = \mathbf{12.58 \text{ Liters}}$$

Summary of Results for Question 7:

a) Pump Flow Rate (Pump ONLY): **Q_p = 24.0 L/min** (Electric Motor: 7.2 kW)

b) With Accumulator:

- Reduced Pump Flow Rate: **Q_p = 9.36 L/min** (Electric Motor: 2.81 kW)
- Required Accumulator Delivery: **ΔV = 2.44 Liters**

• Suitable Accumulator Nominal Size: **$V_0 \approx 12.6$ Liters**

*(Engineering Selection: Commercial size of **16 Liters** or **20 Liters** accumulator shell).*